

DIABETES RISK PREDICTION USING MACHINE LEARNING

PRESENTED BY:

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EXECUTIVE SUMMARY

- Research Focus:
 - Predict diabetes risk using health survey data.
 - Identify key health indicators.
 - Compare multiple machine learning models.
- Key Results:
 - Neural Network and Logistic Regression achieved highest AUC (~0.82). - Class balancing (ROSE) critically improved model fairness.
- Future Recommendations:
 - Expand feature engineering.
 - Deepen model architectures.
 - Validate on newer datasets like BRFSS 2020.

PROJECT TIMELINE

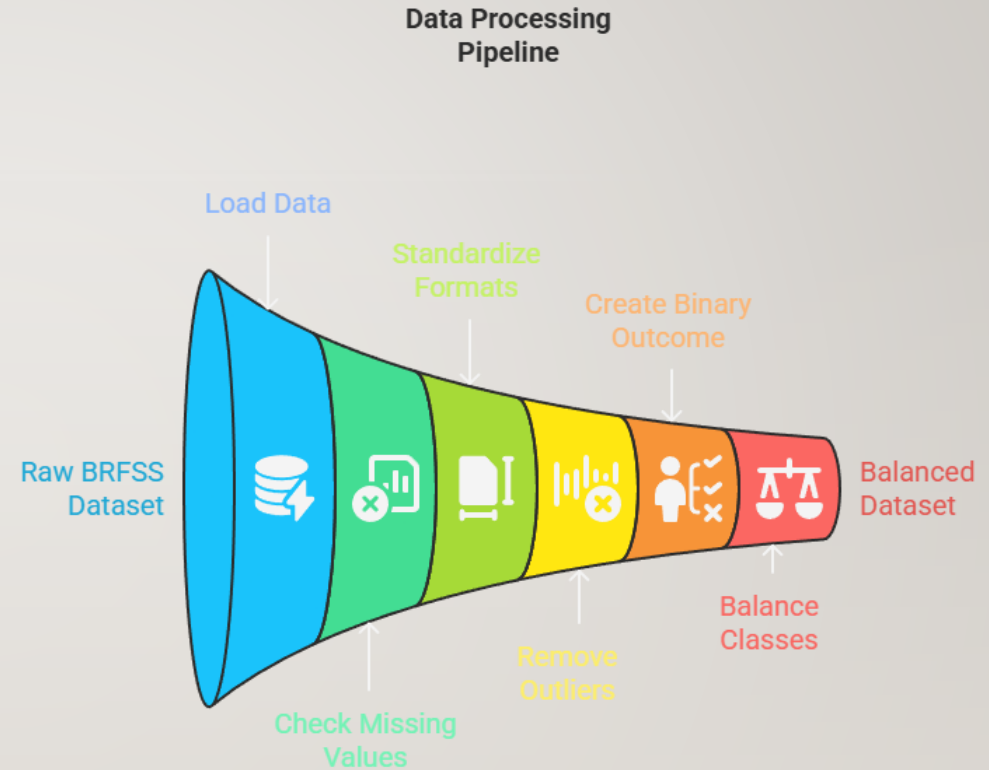
- Timeline:
 - Weeks 1-2: Data Cleaning & EDA
 - Weeks 3-4: Feature Engineering & Balancing
 - Weeks 5-6: Model Training and Evaluation
 - Week 7: Model Tuning and Reporting
- Plan Adjustment:
 - Added ROSE balancing after discovering severe class imbalance.

PROJECT METHODOLOGY

- Approach:
 - 1. Data ingestion and preliminary cleaning.
 - 2. Exploratory data analysis (EDA).
 - 3. Feature extraction and engineering.
 - 4. Class balancing using ROSE.
 - 5. Model building, tuning, evaluation, and comparison.

DATA PROCESSING PIPELINE

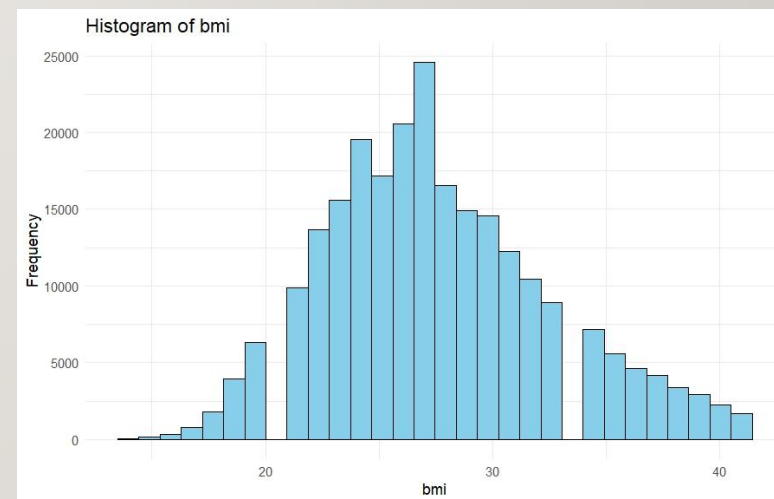
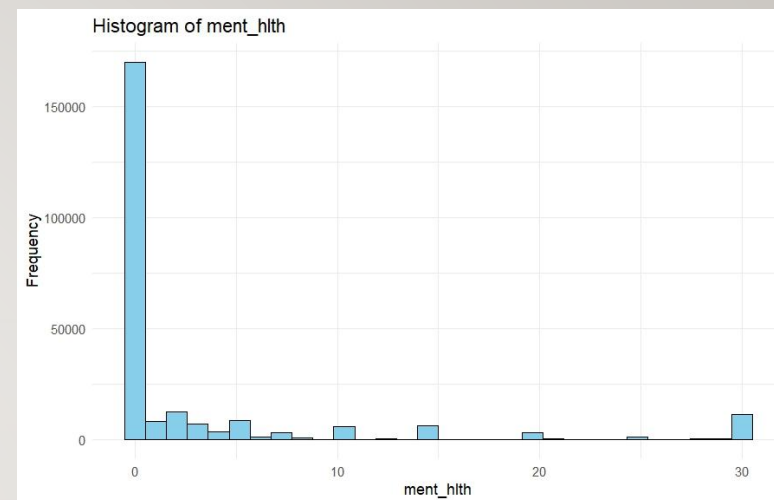
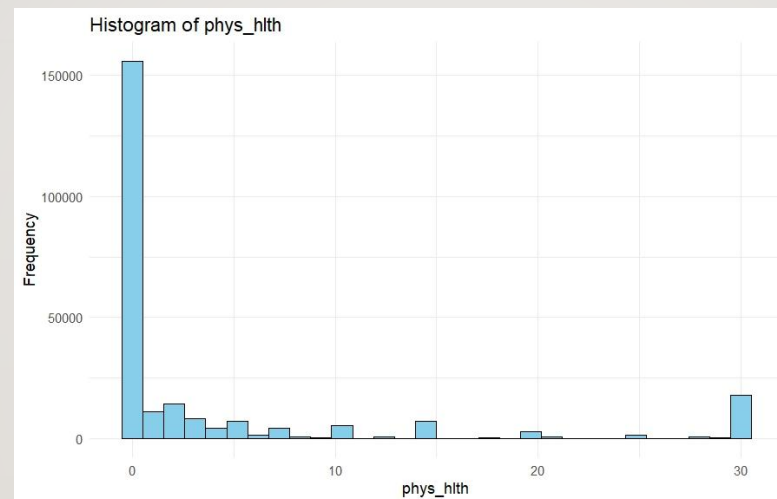
- Loaded BRFSS 2015 dataset (253k records).
- Checked missing values, standardized formats.
- Removed BMI outliers (<13.5 or >41.5).
- Binary outcome (Diabetic vs Non-Diabetic).
- Applied ROSE to balance classes 50/50.



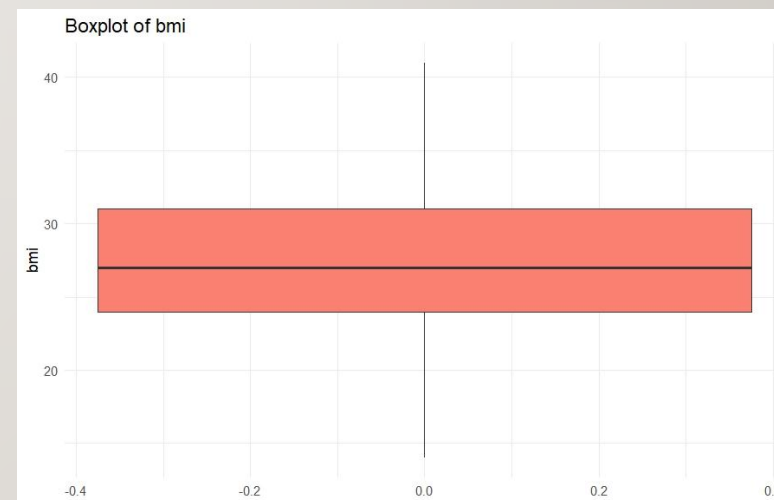
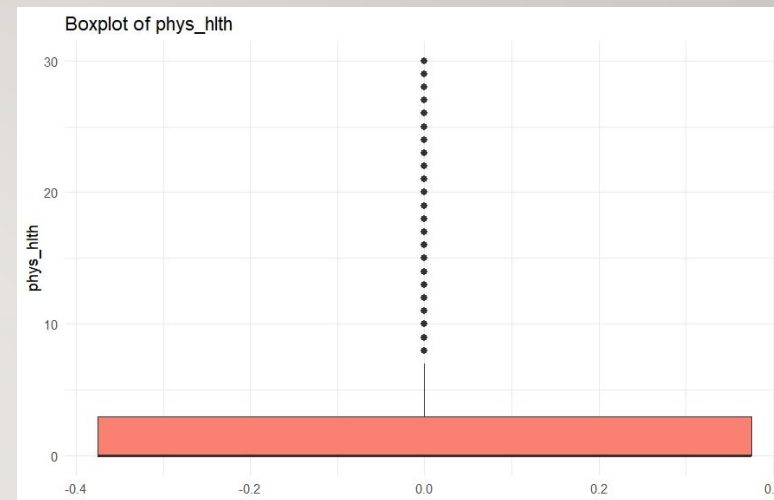
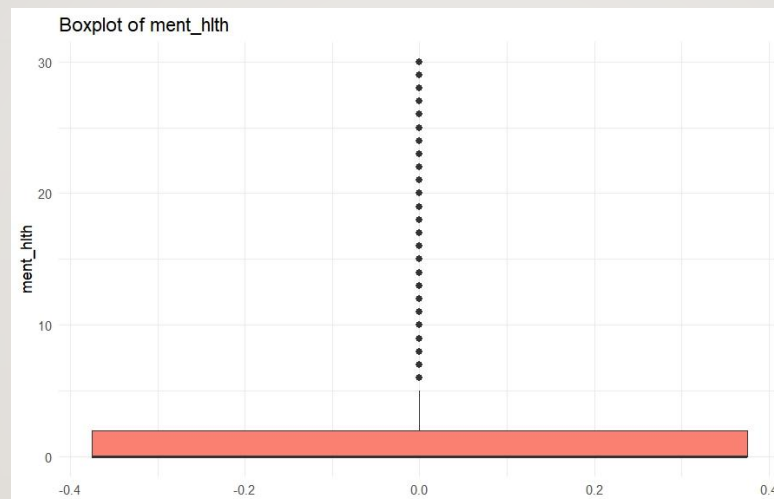
EXPLORATORY DATA ANALYSIS

- Key Insights:
 - BMI mean: 28.38 (overweight range).
 - Mental/Physical Health Days: Heavily right-skewed.
 - BMI strongly correlated with general health.

HISTOGRAMS



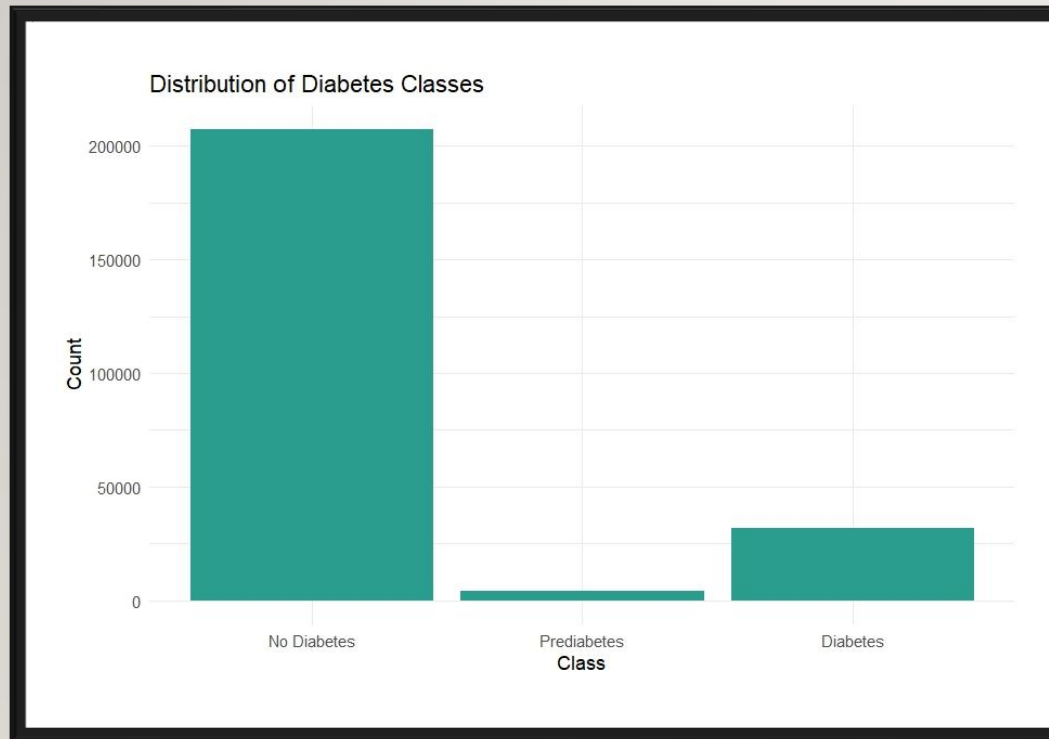
BOXPLOTS



PREPROCESSING AND TRANSFORMATIONS

- Encoded binary variables (e.g., High BP, Cholesterol).
 - Standardized continuous features (BMI, Age, Income, Education).
 - Binary target creation: Diabetic (1) vs Non-Diabetic (0).
- ❖ Note: No unsupervised learning (e.g., clustering, dimensionality reduction) were applied.

CLASS IMBALANCE CORRECTION USING ROSE SAMPLING



- Original Distribution:
 - 90% Non-Diabetic, 8% Diabetic, 2% Prediabetic.
- After ROSE:
 - 97,792 Non-Diabetics
 - 97,276 Diabetics

FEATURE ENGINEERING SUMMARY

- Preserved medically meaningful variables (BMI, Health Days, etc.).
- Encoded binary predictors.
- Standardized continuous predictors for model optimization.

MODEL SELECTION AND EVALUATION CRITERIA

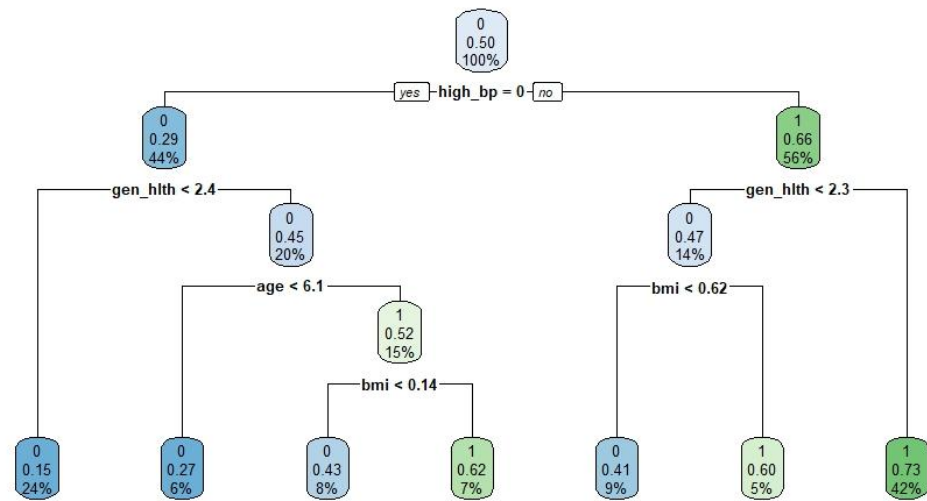
- Models Trained:
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - Neural Network (Tuned with Dropout and Optimizer Tuning)

MODEL TRAINING DETAILS

- Metrics: Accuracy, Precision, Recall, AUC.
- Neural Network Configuration:
 - 64 → 32 neuron layers
 - Dropout: 0.3
 - 80 epochs
 - Adam optimizer with learning rate reduction

DECISION TREE

Decision Tree for Diabetes Prediction



MODEL PERFORMANCE SUMMARY

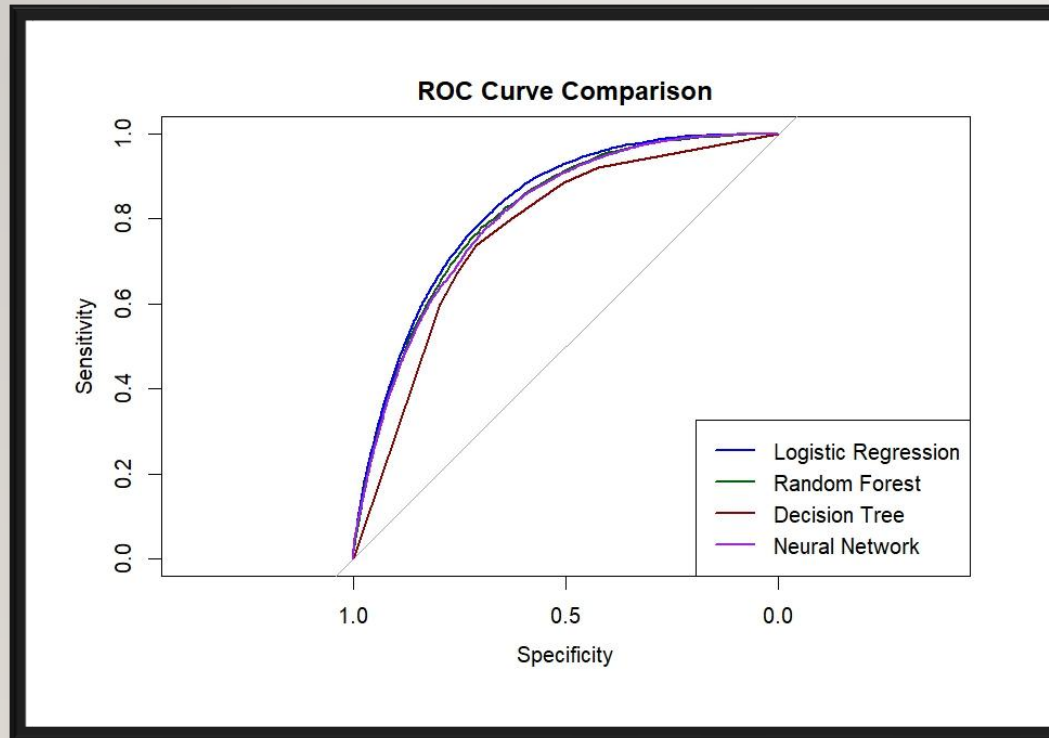
- Highlights:
 - Best recall (76.5%) from Logistic Regression.
 - Highest accuracy (79.3%) and precision (34.1%) from Neural Network.

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	73%	30%	76.5%	0.82
Decision Tree	71.5%	27.8%	73.6%	0.76
Random Forest	78.7%	78.7%	62.5%	0.81
Neural Network	79.3%	34.1%	61.9%	0.817

NEURAL NETWORK VS RANDOM FOREST

- While Random Forest achieved slightly higher overall accuracy (78.7%), Neural Network provided a better balance between Recall (Sensitivity), Precision, and AUC.
- In healthcare risk prediction, high Recall is critical to minimize the chances of missing diabetic cases, even if it slightly lowers overall accuracy.
- Neural Network maintained strong Recall while improving model robustness through regularization (Dropout) and tuning, making it the best trade-off model for deployment.

ROC CURVE COMPARISON

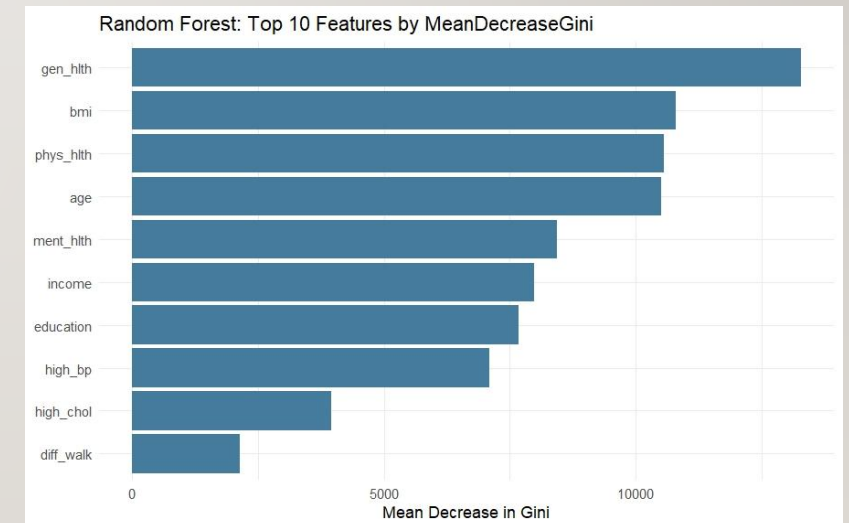
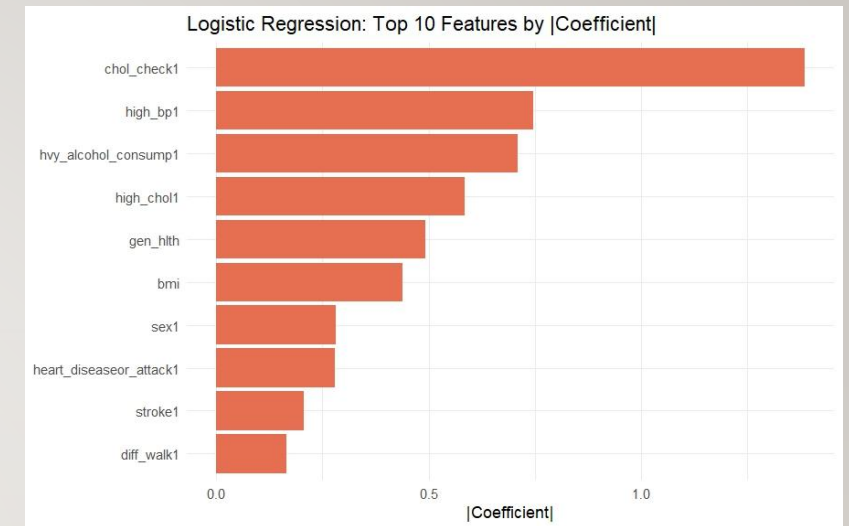


- Neural Network and Logistic Regression close to ideal separation.
- Random Forest strong in overall accuracy, lower in sensitivity.

FEATURE IMPORTANCE INSIGHTS

Top Predictors (Random Forest):

- General Health
- BMI
- Physical Health Days
- Age
- Income
- Education



POSITIVE OUTCOMES

- Neural Network delivered strongest balanced performance.
- Logistic Regression achieved best sensitivity (critical for healthcare).
- Data balancing (ROSE) substantially improved diabetic case detection.

CHALLENGES AND LIMITATIONS

- Precision still moderate (~34%) → risk of false positives.
- Self-reported survey data carries memory and reporting bias.
- Synthetic sampling (ROSE) could create non-natural data patterns.

RISKS & BIASES

- ROSE Limitation: Original class imbalance may still affect results.
- Survey Data: Potential for inaccuracies (recall bias, etc.).
- ROSE Oversampling: Synthetic data might not fully reflect reality.
- NN Overfitting Risk: Early stopping/dropout don't eliminate all risk.
- Recall vs. Precision: Lowering threshold improved recall but hurt accuracy/precision.

FUTURE WORK AND POTENTIAL ENHANCEMENTS

- Apply deeper feature engineering (feature crosses, interactions).
- Hyperparameter optimization (Random Search, Bayesian Tuning).
- Validate models on external datasets like BRFSS 2020 or PIMA.

FINAL TAKEAWAYS

- Key Learnings:
 - Data quality and class balancing are foundational for predictive success.
 - Machine Learning models can effectively predict diabetes risks.
 - Rigorous external validation is needed before real-world deployment.

THANK YOU

THANK YOU FOR YOUR TIME!

